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Research Proposal

Aspect-Based Sentiment Analysis and Contextual Intent Extraction from User-Generated Banking Customer Reviews

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Declaration

I, Rotondwa Mavhengani, declare that this research proposal titled *"Aspect-Based Sentiment Analysis and Contextual Intent Extraction from User-Generated Banking Customer Reviews"* is my own original work and has not previously been submitted, in whole or in part, for any degree or examination at this or any other university.

I further declare that all sources used or referred to in this proposal have been properly acknowledged. Any assistance received in preparing the proposal has also been appropriately acknowledged.

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Abstract

Banks receive substantial volumes of customer feedback through social media, online review platforms, and public complaint websites. Such feedback contains information about transactions, mobile and online banking, cards, accounts, fees, fraud, branch service, and customer support. However, a single review may discuss several banking aspects and express different sentiments toward them. Document-level sentiment analysis is therefore insufficient because it assigns one overall polarity without identifying the aspect responsible for the opinion (Hu and Liu, 2004; Pontiki et al., 2014).

This study proposes a banking-domain adaptation of Transformer-Based Aspect-Level Sentiment Classification with Intent (TASCI), introduced by Chaudhry et al. (2025). The proposed pipeline has three stages. First, a self-attention token classifier will extract explicit banking aspect terms using query, key, and value projections, scaled dot-product attention, and multi-head attention. Second, a gated recurrent unit (GRU)-based sequence-to-sequence model will predict customer intent from the current sentence and, where available, its preceding sentence or conversational context. Third, extracted aspects will be represented using GloVe embeddings, processed by a Transformer encoder, combined with the predicted intent vector, and classified as positive, negative, or neutral.

The study will use international Kaggle banking reviews, a South African banking-tweet dataset, and a permitted sample of South African Hellopeter banking reviews. The data will be manually annotated for aspect spans, aspect-level sentiment, and intent categories including complaint, compliment, request for assistance, suggestion, warning, churn signal, and general feedback. The adapted TASCI model will be compared with Support Vector Machine, BiLSTM, and BERT baselines. Evaluation will include token- and span-level F1 for aspect extraction, macro-F1 for intent and sentiment classification, end-to-end aspect–sentiment performance, and ablation experiments with and without intent integration. The study will determine whether contextual intent improves aspect-level sentiment analysis for international and South African banking feedback.

1. Chapter One: Introduction and Background

1.1 Background of the Study

Customer feedback enables banks to understand experiences involving transaction reliability, account access, customer support, cards, fees, fraud handling, digital platforms, and branch services. User-generated reviews are particularly valuable because they contain direct descriptions of service successes and failures (Kazmaier and Van Vuuren, 2020; Semenya, 2024). South African industry studies also show that customers increasingly use public channels to express compliments, complaints, and operational frustrations concerning local banks (DataEQ, 2023; DataEQ, 2025).

Traditional sentiment analysis usually assigns one positive, negative, or neutral label to an entire text. This is inadequate when a review contains several aspects with different polarities. ABSA addresses this limitation by extracting an aspect and determining the sentiment directed toward it (Hu and Liu, 2004; Pontiki et al., 2014). Banking reviews require such fine-grained analysis because one message may praise a support agent while criticising a delayed payment or failed application.

ABSA research remains concentrated in restaurant, laptop, hotel, and e-commerce datasets, producing a domain-transfer problem for banking terminology and service structures (Hua et al., 2024; Chebolu et al., 2023). Banking customer feedback includes specialised concepts such as blocked cards, unauthorised transactions, account verification, payment reversals, fraud investigations, and complaint escalation (Ogunleye, 2021; Kazmaier and Van Vuuren, 2020).

Intent provides a further layer of interpretation. Two reviews can express equally negative sentiment but communicate different goals: one may complain, another may request assistance, and another may indicate an intention to close an account. Chaudhry et al. (2025) integrated aspect extraction, contextual intent analysis, and aspect-level sentiment classification in TASCI. Their ablation study showed lower performance when intent analysis was removed, indicating that intent can contribute useful contextual information to sentiment classification.

1.2 Problem Statement

Existing document-level sentiment systems cannot adequately explain which banking service aspect caused a customer's opinion. Furthermore, aspect-only systems may identify what the customer discusses and how the customer feels, but not what the customer is trying to achieve. This limits the operational usefulness of automatic feedback analysis for complaint routing, assistance prioritisation, warning detection, and churn-risk identification.

A second challenge is domain imbalance. Most benchmark ABSA datasets are based on restaurants and laptops rather than financial services (Pontiki et al., 2014; Hua et al., 2024; Chebolu et al., 2023). Models trained in these domains may not transfer reliably to banking concepts and informal customer-service messages.

A third challenge concerns contextual availability. TASCI predicts intent by relating a current sentence to a preceding sentence (Chaudhry et al., 2025). Some banking reviews are multi-sentence texts or connected conversations, while many tweets are isolated records. A banking adaptation must therefore distinguish between context-available and context-unavailable cases rather than assuming that every record has a valid previous sentence.

Finally, international datasets do not fully represent South African banking conditions. This study will combine international banking reviews with South African tweets and permitted Hellopeter samples to evaluate both broader patterns and locally relevant feedback.

1.3 Research Aim

The main aim is to develop and evaluate a banking-domain adaptation of TASCI for aspect extraction, contextual intent extraction, and intent-aware aspect-level sentiment classification from user-generated banking customer reviews.

1.4 Research Objectives

1. Identify major banking service aspects in international and South African customer feedback.
2. Develop a self-attention token-classification model for extracting explicit banking aspect terms.
3. Develop a GRU-based sequence-to-sequence model for predicting customer intent from current and available preceding context.
4. Classify sentiment for each extracted aspect by integrating GloVe aspect representations, Transformer-generated contextual representations, and the predicted intent vector.
5. Develop and validate a banking-specific annotation scheme for aspect spans, aspect-level sentiment, and customer intent.
6. Compare the adapted TASCI model with SVM, BiLSTM, and BERT baselines.
7. Conduct ablation experiments to measure the contribution of contextual intent and other pipeline components.

8. Compare international and South African aspect, sentiment, and intent patterns.
9. Derive practical insights into complaints, assistance requests, compliments, warnings, and churn signals.

1.5 Research Questions

1. What banking aspects are most frequently expressed in user-generated customer feedback?
2. How effectively can self-attention-based sequence labelling extract banking aspect terms?
3. How effectively can a GRU-based model identify customer intent from current and available preceding context?
4. Does integrating predicted customer intent improve aspect-level sentiment classification?
5. How does the adapted TASCI model compare with SVM, BiLSTM, and BERT baselines?
6. Which architecture components contribute most to end-to-end performance?
7. What similarities and differences exist between international and South African banking feedback?
8. How can the extracted aspect, sentiment, and intent information support banking customer-experience analysis?

1.6 Significance of the Study

The study contributes methodologically by adapting an integrated intent-aware ABSA architecture to a domain not evaluated in the original TASCI study. It contributes empirically through a banking-specific annotated dataset that includes a South African component. Practically, the framework may help distinguish the service area being discussed, the attitude expressed toward it, and the action implied by the message.

1.7 Scope of the Study

The study will analyse English-language, text-based banking reviews and tweets from international Kaggle sources, South African banking tweets, and permitted Hellopeter samples. It will not analyse images, audio, private complaint records, or personally identifiable banking information. The study will focus on explicit aspect terms and the intent categories defined in the annotation scheme.

2. Chapter Two: Literature Review

2.1 Conceptual Foundations of ABSA

ABSA identifies specific targets or aspects and assigns sentiment to each rather than producing one label for an entire review (Hu and Liu, 2004; Pontiki et al., 2014). Its subtasks include aspect-term extraction, aspect categorisation, opinion extraction, polarity classification, and aspect–opinion association. Chebolu et al. (2023) extend this structure through opinion–aspect–target–sentiment quadruple extraction, demonstrating the value of modelling related opinion elements together.

2.2 Statistical, Sequential, and Transformer Approaches

Traditional ABSA uses lexicons, rules, dependency features, TF–IDF, and classifiers such as SVM and Naive Bayes. These remain useful baselines when labelled data is limited (Ogunleye, 2021). Sequential neural models such as LSTM and BiLSTM improve contextual representation by processing ordered token sequences. Attention mechanisms further allow a model to emphasise tokens relevant to a selected aspect (Feng et al., 2022; Jayakody et al., 2024).

Transformers use self-attention to model long-range token relationships and contextual meaning (Vaswani et al., 2017). BERT-based and related approaches have achieved strong ABSA performance, although results remain dependent on domain, annotation quality, and dataset design (Hua et al., 2024; Jayakody et al., 2024).

2.3 ABSA in Banking and South African Customer Feedback

Banking ABSA remains less developed than restaurant and electronics ABSA. Existing financial-service studies demonstrate that machine-learning sentiment models can extract useful information from bank feedback, but many remain document-level or channel-specific (Ogunleye, 2021; Kazmaier and Van Vuuren, 2020). Semenya (2024) analysed Hellopeter reviews for South African retail banks, confirming the relevance of public review data for local customer-experience research.

2.4 Contextual Intent and TASCI

Chaudhry et al. (2025) proposed TASCI, an integrated architecture containing self-attention aspect extraction, GRU-based contextual intent prediction, and Transformer-based aspect-level sentiment classification. In its aspect extractor, token embeddings are projected to query, key,

and value vectors; scaled attention scores are normalised into weights; and multi-head attention refines contextual token representations. A token classifier then assigns aspect labels.

The intent component encodes the previous and current sentences with GRUs. Attention identifies the parts of the previous sentence most relevant to the current sentence, producing a context vector used by the decoder to predict intent. In the final stage, GloVe aspect embeddings are processed through a Transformer encoder and combined with the predicted intent vector before sentiment classification (Chaudhry et al., 2025; Pennington et al., 2014).

TASCI achieved 89.10% accuracy and 83.38% macro-F1 on Restaurant, 84.81% accuracy and 78.63% macro-F1 on Laptop, and 79.08% accuracy and 77.27% macro-F1 on Twitter. Its ablation study showed reduced performance without intent analysis (Chaudhry et al., 2025). However, it was not evaluated on banking feedback, and its contextual intent method assumes that preceding discourse is available.

3. Chapter Three: Critical Review and Research Gap

3.1 Critical Review of Foundational and Banking Studies

Hu and Liu (2004) and Pontiki et al. (2014) established feature-level and benchmark ABSA, but their product and restaurant domains do not represent banking-specific language. Banking sentiment studies demonstrate domain relevance, yet many do not jointly model explicit aspect spans, aspect-level polarity, and customer intent (Ogunleye, 2021; Kazmaier and Van Vuuren, 2020; Semenya, 2024).

3.2 Critical Review of TASCI

TASCI's strength is its integration of three related tasks rather than treating intent as an unrelated output. Its results and ablation experiments provide evidence that contextual intent may improve aspect-level sentiment classification (Chaudhry et al., 2025). Nevertheless, three limitations matter for the proposed study. First, TASCI was not evaluated on banking data. Second, its GRU intent extractor expects preceding context, which may be absent in isolated tweets. Third, its multi-component architecture requires reliable manual annotation and careful error analysis.

3.3 Research Gap

There is limited research adapting an intent-aware ABSA pipeline to banking customer feedback, particularly a dataset containing both international and South African sources. There is also limited evidence on whether contextual intent improves sentiment classification when only some records contain valid preceding context. This study addresses the gap by adapting TASCI to banking, introducing a context-availability strategy, and evaluating the full system against no-intent and current-sentence-only alternatives.

4. Chapter Four: Research Methodology

4.1 Research Design

The study will use an experimental, comparative design. It will construct and annotate a banking corpus, train the proposed three-stage model, compare it with baseline systems, and conduct ablation experiments. The independent experimental conditions will include model type and context availability, while the dependent variables will be aspect extraction, intent classification, sentiment classification, and end-to-end performance.

4.2 Data Sources and Sampling

The corpus will combine: (1) international banking tweets and reviews from Kaggle; (2) the *Tweets of Popular South African Banks* dataset; and (3) an ethically permitted sample of Hellopeter reviews. Records will be filtered for meaningful banking feedback. Stratified sampling will be used where possible to balance source, region, and preliminary aspect coverage. The annotated subset will be sized according to available time and annotator capacity rather than assuming that all millions of source records can be labelled.

4.3 Preprocessing and Context Construction

Duplicate and near-duplicate records, empty texts, irrelevant posts, URLs, and direct identifiers will be removed or anonymised. Negation, intensifiers, emojis with semantic value, and meaningful hashtags will be retained. Reviews will be segmented into sentences and tokenised.

For a multi-sentence review, the previous sentence will serve as S_{t-1} and the current sentence as S_t. For connected social-media conversations, a preceding post will be used only when its text and relationship are available. Isolated records will receive a special `[NO_CONTEXT]` representation. Results will be reported separately for genuine-context and no-context subsets.

4.4 Annotation Scheme

The corpus will be annotated at two levels. At token level, explicit aspect spans will use BIO labels: `B-ASP`, `I-ASP`, and `O`. Each extracted span will also receive an aspect category and polarity label. At sentence or review level, one dominant intent will be assigned.

Table 1: Proposed banking annotation scheme

Dimension	**Labels**	**Purpose**
Aspect span	`B-ASP`, `I-ASP`, `O`	Identifies the exact tokens forming an explicit banking aspect.
Aspect category	Customer support; digital banking; transactions; cards; accounts; fees; fraud/security; branch service; general service	Groups semantically related aspect terms into banking service areas.
Aspect sentiment	Positive; negative; neutral	Assigns polarity separately to every annotated aspect span.
Customer intent	Complaint; compliment; request for assistance; suggestion; warning; churn signal; general feedback	Identifies the dominant communicative purpose of a sentence or review.
Context status	Previous context available; no context available	Supports separate evaluation of contextual and isolated records.

Two annotators will independently label a pilot sample. Cohen's Kappa will assess categorical agreement for sentiment, intent, and aspect category. Exact-span agreement and token-level F1 will assess aspect-boundary consistency. Disagreements will be adjudicated and the guidelines revised before full annotation.

4.5 Proposed TASCI-Style Pipeline

4.5.1 Stage One: Self-Attention Aspect Extraction

For token sequence $X = [x_1, \ldots, x_n]$, token embeddings will be projected into query, key, and value vectors:

$$Q_i = W^Q x_i, \qquad K_i = W^K x_i, \qquad V_i = W^V x_i. \tag{1}$$

Scaled dot-product attention and normalised weights will be calculated as:

$$\text{score}_{ij} = \frac{Q_i K_j^T}{\sqrt{d_k}}, \tag{2}$$

$$\alpha_{ij} = \frac{\exp(\text{score}_{ij})}{\sum_{k=1}^{n} \exp(\text{score}_{ik})}, \tag{3}$$

$$z_i = \sum_{j=1}^{n} \alpha_{ij} V_j. \tag{4}$$

Multi-head attention will concatenate head-specific outputs:

$$Z_i = \text{Concat}(z_i^1, \ldots, z_i^h)W^O. \tag{5}$$

A linear softmax classifier will predict the BIO label for each token. Cross-entropy loss will train the extractor (Vaswani et al., 2017; Chaudhry et al., 2025).

4.5.2 Stage Two: GRU-Based Contextual Intent Extraction

Separate GRU encoders will process S_{t-1} and S_t. Attention over previous-sentence hidden states will produce a contextual summary:

$$c_t = \sum_{i=1}^{m} \alpha_i h_{t-1}^{(i)}. \tag{6}$$

The context vector and the current sentence's final hidden state will be combined:

$$d_t = \tanh\left(W^d[c_t; h_t^{(\text{final})}]\right), \tag{7}$$

and intent will be predicted as:

$$\hat{I}_t = \text{softmax}(W^o d_t + b^o). \tag{8}$$

For isolated records, the no-context representation will replace c_t. Cross-entropy loss will be used for intent classification (Cho et al., 2014; Chaudhry et al., 2025).

4.5.3 Stage Three: Transformer Sentiment Classification with Intent

For aspect A, GloVe token embeddings will be aggregated as:

$$v_A = \frac{1}{|A|} \sum_{i \in A} e_i. \tag{9}$$

A Transformer encoder will produce contextual aspect representation H_A. The sentiment classifier will concatenate H_A with predicted intent vector I_t:

$$s_A = \text{softmax}\left(W^S[H_A; I_t] + b^S\right). \tag{10}$$

Concatenation is used to avoid assuming identical aspect and intent dimensions. The model will predict positive, negative, or neutral sentiment separately for each aspect (Pennington et al.,

2014; Chaudhry et al., 2025).

4.6 Baseline Models

The proposed system will be compared with: (1) SVM using TF–IDF features; (2) BiLSTM; and (3) BERT. These provide statistical, sequential, and contextual Transformer baselines while keeping the project feasible within the three-month period. All models will use the same training, validation, and test partitions.

4.7 Ablation Study

1. Full proposed model.
2. Model without intent integration.
3. Intent model using current sentence only.
4. Intent model using previous and current context.
5. Model without the Transformer sentiment encoder.
6. Gold-aspect sentiment evaluation.
7. Predicted-aspect end-to-end evaluation.

The full-versus-no-intent comparison will directly test whether intent improves banking aspect-level sentiment classification.

4.8 Evaluation Strategy

Aspect extraction will use token-level and exact-span precision, recall, and F1. Intent and sentiment classification will use accuracy, macro-precision, macro-recall, macro-F1, weighted-F1, and confusion matrices. End-to-end evaluation will require both the aspect span and its sentiment to be correct. A complete tuple measure will additionally assess aspect, sentiment, and intent together.

Data will be divided into training, validation, and test sets using stratification where possible. Duplicate reviews and posts from the same identifiable thread will remain in one split to reduce leakage. Results will be reported jointly and separately for international, South African, context-available, and no-context subsets.

4.9 Ethical Considerations

Usernames, names, account references, contact information, and other identifiers will be removed. Hellopeter data will only be used if collection is compatible with platform terms and university ethics requirements. The study will not attempt to identify customers, employees, or private banking cases. Direct quotations will be anonymised or paraphrased where necessary.

4.10 Limitations

The mixed sources may differ in writing style, time period, geography, and platform norms. Some intent classes may be rare, and not all records contain preceding context. Public posts may not represent all banking customers. The integrated model also requires more annotation and computation than a document-level classifier. These limitations will be addressed through source-specific reporting, class-sensitive metrics, ablation experiments, and transparent context-status labels.

5. Chapter Five: Project Plan, Expected Results and Conclusion

5.1 Research Timeline

Table 2: Three-month research timeline

Month	Main activities
Month 1	Literature review; dataset acquisition; ethics and platform checks; cleaning; context construction; annotation guideline design; pilot annotation.
Month 2	Full annotation; self-attention aspect extractor; GRU intent model; SVM, BiLSTM, and BERT baselines; Transformer sentiment integration.
Month 3	Training; component evaluation; end-to-end evaluation; ablation experiments; regional comparison; analysis; writing and revision.

5.2 Gantt-Style Work Plan

Table 3: Gantt-style weekly work plan

Activity	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Literature review	X	X	X	X								
Dataset and ethics preparation	X	X	X									
Cleaning and context construction		X	X	X								
Guidelines and pilot annotation			X	X	X							
Full annotation					X	X	X					
Aspect extractor						X	X					
GRU intent model						X	X	X				
Transformer sentiment integration							X	X				
Baseline development						X	X	X				
Evaluation and ablation								X	X	X		
Analysis and writing				X	X	X	X	X	X	X	X	
Final revision and submission											X	X

5.3 Expected Results

The study is expected to produce a banking-domain adaptation of TASCI and a corresponding annotation scheme. TASCI's published benchmark performance suggests that intent integration

can improve aspect-level sentiment classification, but equivalent banking performance cannot be assumed because banking terminology, public complaints, and context availability differ from the original datasets (Chaudhry et al., 2025).

The full model is expected to outperform the no-intent version when genuine preceding context is available. The improvement may be strongest for operationally distinctive intents such as assistance requests, warnings, and churn signals. Isolated tweets may show a smaller intent benefit, which will be examined through the context-availability experiment.

5.4 Expected Contributions

1. A banking-specific annotated corpus with South African and international components.
2. A banking-domain adaptation of the TASCI pipeline.
3. Evidence on whether contextual intent improves banking aspect-level sentiment classification.
4. An evaluation strategy separating gold-aspect, predicted-aspect, and end-to-end performance.
5. Practical insight into banking aspects, sentiment, complaints, assistance requests, warnings, and churn signals.

5.5 Conclusion

This proposal positions contextual intent as information that may improve both the accuracy and operational usefulness of banking ABSA. The proposed model will extract aspect terms with self-attention, infer intent with a GRU-based contextual model, and classify aspect sentiment by combining GloVe–Transformer aspect representations with the predicted intent vector. Through baseline comparison, context-aware evaluation, and ablation analysis, the study will determine whether the TASCI architecture transfers effectively to international and South African banking customer feedback.

References

1. Bahdanau, D., Cho, K. and Bengio, Y. (2014). Neural machine translation by jointly learning to align and translate. *arXiv preprint arXiv:1409.0473*.

2. Batoog. (n.d.). *Bank customer tweets (10000)*. Kaggle dataset. Available at: https://www.kaggle.com/datasets/batoog/bank-customer-tweets-10000.

3. Blomerus, N. (n.d.). *Tweets of Popular South African Banks*. Kaggle dataset. Available at: https://www.kaggle.com/datasets/nicholasblomerus/twitter-analysis-of-popular-south-a

4. Chaudhry, H. N., Kulsoom, F., Ullah Khan, Z., Aman, M., Khan, S. U. and Albanyan, A. (2025). TASCI: transformers for aspect-based sentiment analysis with contextual intent integration. *PeerJ Computer Science*, 11, e2760. https://doi.org/10.7717/peerj-cs.2760.

5. Chebolu, S., Raman, R. and Srinivasan, K. (2023). OATS: Opinion Aspect Target Sentiment Quadruple Extraction Dataset for Aspect-Based Sentiment Analysis. *arXiv preprint arXiv:2308.14266*.

6. Cho, K., Van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H. and Bengio, Y. (2014). Learning phrase representations using RNN encoder–decoder for statistical machine translation. *arXiv preprint arXiv:1406.1078*.

7. DataEQ. (2023). *FNB takes top spot, banking SA's most loved industry*. Available at: https://dataeq.com/resources/news/sa-banking-sentiment-index-fnb-takes-top-spot.

8. DataEQ. (2025). *South African Banking Sentiment Index*. Available at: https://dataeq.com/resources/reports/sa-banking-sentiment-index/.

9. Feng, S., Qu, G., Baral, S. and Li, W. (2022). Unrestricted Attention May Not Be All You Need: Masked Attention Mechanism Focuses Better on Relevant Parts in Aspect-Based Sentiment Analysis. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*.

10. Hu, M. and Liu, B. (2004). Mining and summarizing customer reviews. In *Proceedings of the Tenth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 168–177.

11. Hua, Y. C., Denny, P., Wicker, J. and Taskova, K. (2024). A systematic review of

aspect-based sentiment analysis: domains, methods and trends. *Artificial Intelligence Review*.

12. Jayakody, D. et al. (2024). Aspect-based sentiment analysis techniques: A comparative study. *arXiv preprint arXiv:2407.02834*.

13. Kazmaier, J. and Van Vuuren, J. H. (2020). Sentiment analysis of unstructured customer feedback for a retail bank. *ORiON*, 36(1), pp. 35–71.

14. Ogunleye, B. (2021). *Statistical Learning Approaches to Sentiment Analysis in the Nigerian Banking Context*. Doctoral dissertation.

15. Pennington, J., Socher, R. and Manning, C. D. (2014). GloVe: Global vectors for word representation. In *Proceedings of EMNLP 2014*, pp. 1532–1543.

16. Pontiki, M., Galanis, D., Pavlopoulos, J., Papageorgiou, H., Androutsopoulos, I. and Manandhar, S. (2014). SemEval-2014 task 4: Aspect based sentiment analysis. In *Proceedings of SemEval 2014*, pp. 27–35.

17. Semenya, R. H. (2024). *Sentiment and thematic analysis of customer experience and satisfaction with South African retail banks*. Master's dissertation, University of Johannesburg.

18. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L. and Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems*, 30.